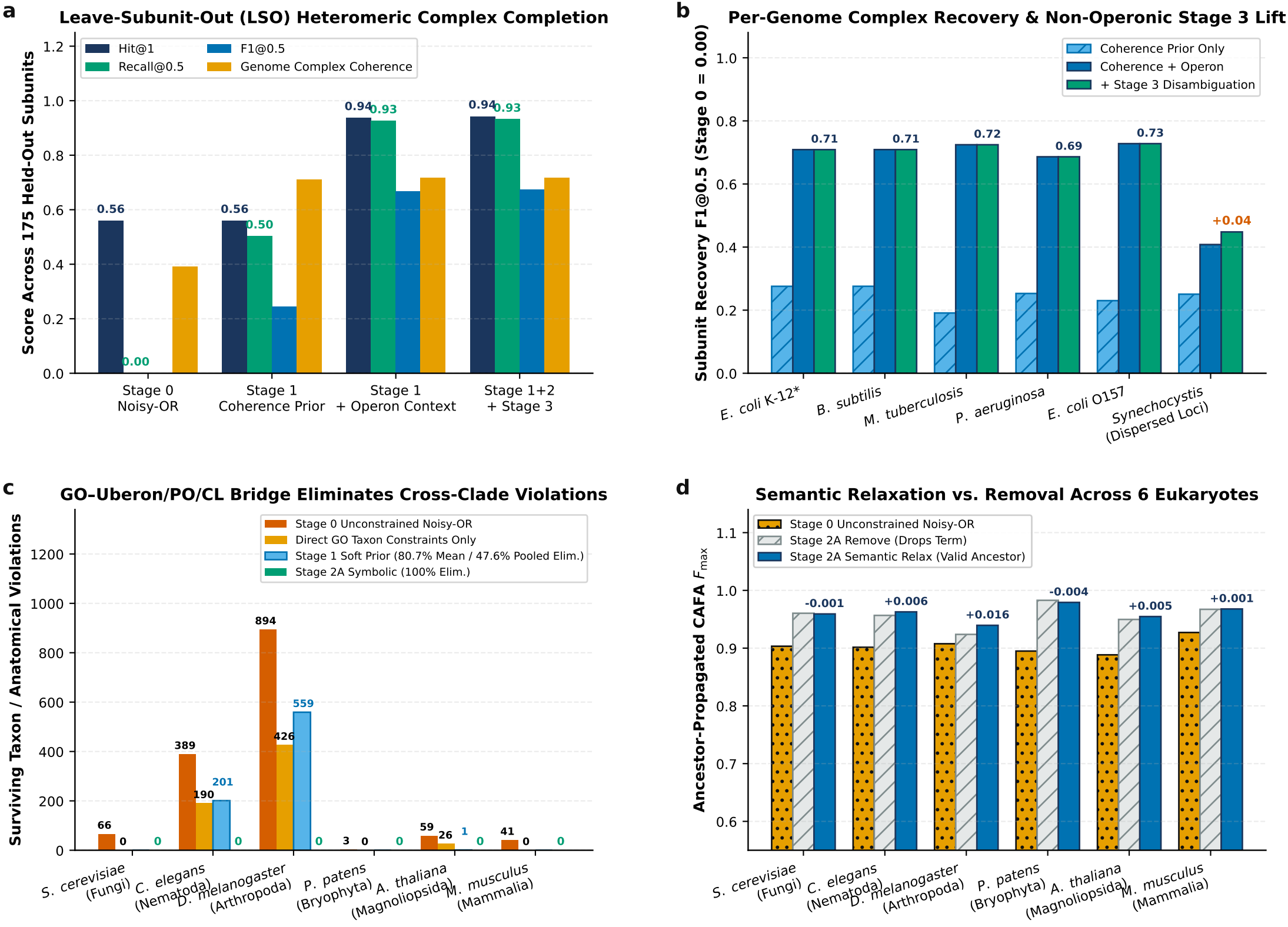

a
Leave-Subunit-Out (LSO) Heteromeric Complex Completion
Score Across 175 Held-Out Subunits
Hit@1
Recall@0.5
F1@0.5
Genome Complex Coherence
0.56
0.00
0.56
0.50
0.94
0.93
0.94
0.93
Stage 0
Noisy-OR
Stage 1
Coherence Prior
Stage 1
+ Operon Context
Stage 1+2
+ Stage 3
b
Per-Genome Complex Recovery & Non-Operonic Stage 3 Lift
Subunit Recovery F1@0.5 (Stage 0 = 0.00)
Coherence Prior Only
Coherence + Operon
+ Stage 3 Disambiguation
0.71
0.71
0.72
0.69
0.73
+0.04
E. coli K-12*
B. subtilis
M. tuberculosis
P. aeruginosa
E. coli O157
Synechocystis
(Dispersed Loci)
c
GO-Uberon/PO/CL Bridge Eliminates Cross-Clade Violations
Surviving Taxon / Anatomical Violations
Stage 0 Unconstrained Noisy-OR
Direct GO Taxon Constraints Only
Stage 1 Soft Prior (80.7% Mean / 47.6% Pooled Elim.)
Stage 2A Symbolic (100% Elim.)
66
0
0
389
190
201
0
894
426
559
0
3
0
0
59
26
1
0
41
0
0
S. cerevisiae
(Fungi)
C. elegans
(Nematoda)
D. melanogaster
(Arthropoda)
P. patens
(Bryophyta)
A. thaliana
(Magnoliopsida)
M. musculus
(Mammalia)
d
Semantic Relaxation vs. Removal Across 6 Eukaryotes
Ancestor-Propagated CAFA F_{max}
Stage 0 Unconstrained Noisy-OR
Stage 2A Remove (Drops Term)
Stage 2A Semantic Relax (Valid Ancestor)
-0.001
+0.006
+0.016
-0.004
+0.005
+0.001
S. cerevisiae
(Fungi)
C. elegans
(Nematoda)
D. melanogaster
(Arthropoda)
P. patens
(Bryophyta)
A. thaliana
(Magnoliopsida)
M. musculus
(Mammalia)